

Part 2: Face Morphing Attack Detection

1. Chosen Article

Venkatesh, S.; Raghavendra, R.; Raja, K.; Busch, C. "Single Image Face Morphing Attack Detection Using Ensemble of Features." Proc. IEEE 23rd International Conference on Information Fusion (FUSION), 2020. This is a Single-image Morphing Attack Detection (S-MAD) method: it decides bona fide vs. morphed from one image alone, with no trusted live-capture reference -- the right fit for our setup, since Part 1 only produced the two source photos that were morphed together, not a separate trusted capture of either subject.

2. Method (as implemented)

Each face image is represented in two color spaces (YCbCr, HSV, 6 channel images total). Each channel gets a 3-level Laplacian pyramid to expose morphing residue at multiple scales (18 scale-space images per face). Three texture descriptors are computed on every scale-space image and concatenated per descriptor type across all 18: Local Binary Patterns (LBP, block-wise), Histogram of Oriented Gradients (HOG), and Binarized Statistical Image Features (BSIF). Each of the 3 resulting streams is classified independently by a Collaborative Representation Classifier (CRC): a probe vector is reconstructed as a ridge-regularized combination of each class's training vectors, and the class with lower reconstruction residual wins. The per-stream scores are z-score normalized against their own training distribution, then fused by summation.

src/mad_features.py, src/mad_crc.py -- pipeline core:

```
def scale_space_images(img_bgr): # 6 color channels x 3 Laplacian levels
    return [_resize(level) for ch in to_color_channels(img_bgr)
            for level in laplacian_pyramid(ch)]

class CRCClassifier:
    def fit(self, X, y): # dictionary D_c = training vectors of class c
        D = Xn[y == c].T; self.proj_[c] = inv(D.T @ D + lam*I) @ D.T
    def score(self, X): # residual(bona fide) - residual(morph)
        return [self._residual(x, 0) - self._residual(x, 1) for x in Xn]
```

3. Adaptations from the Paper

- BSIF normally uses a filter bank pre-learned via ICA on ~50,000 natural-image patches (Kannala & Rahtu, 2012). We do not have that corpus, so we learn a smaller ICA filter bank (6 filters, 5x5) directly from each cross-validation fold's own bona fide training images instead -- same principle, much smaller and fold-local training data.
- The paper evaluates digital and print-scanned morphs on 1,309 bona fide / 2,608 morphed images with a single fixed identity-disjoint train/test split. Our Part 1 dataset has 116 bona fide / 58 morphed digital images only, so we use 5-fold pair-disjoint cross-validation (GroupKFold on the source-pair id) to get an out-of-fold score for every image while still preventing any image pair from straddling train and test.
- We z-score each descriptor stream's scores against its own training distribution before the sum-rule fusion; the original paper does not state whether it normalizes before summing, and skipping this step let BSIF's very high-dimensional (73,728-d) residual scale dominate the fused score in our own initial run.

4. Dataset and Evaluation Protocol

Bona fide class: the 116 original LFW source images (Subject A and B) from Part 1's 58 pairs. Attack class: the 58 corresponding alpha=0.5 morphs. Evaluated with 5-fold pair-disjoint cross-validation and reported with the ISO/IEC 30107-3 metrics: APCER (attacks classified as bona fide), BPCER (bona fide classified as attacks), and D-EER (operating point where the two are equal).

5. Results

Configuration	D-EER (%)
LBP stream alone	18.97
HOG stream alone	18.97
BSIF stream alone	43.10
Fused ensemble (LBP+HOG+BSIF)	17.24

At the ISO/IEC 30107-3 operating points: BPCER = 57.76% at APCER = 5%, and BPCER = 41.38% at APCER = 10%.

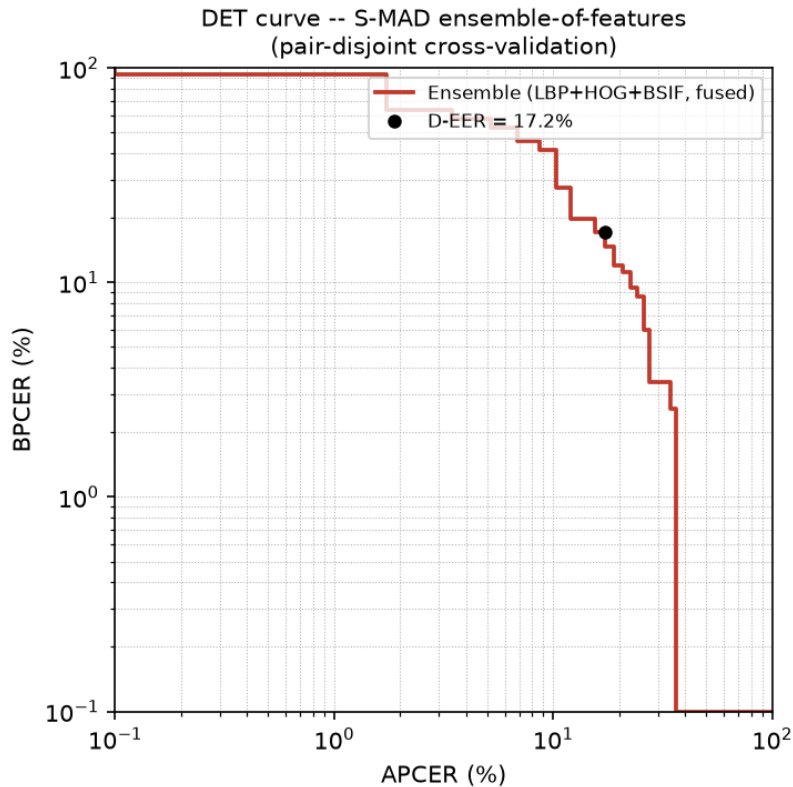


Figure 1: DET curve of the fused ensemble score, pair-disjoint 5-fold cross-validation (both axes log-scaled per ISO/IEC 30107-3 convention).

6. Observations and Limitations

- The fused ensemble (D-EER 17.2%) is noticeably weaker than the paper's own reported 5.99-5.64% D-EER, mainly because our training set is two orders of magnitude smaller (174 vs 3,900+ images) and identity-disjoint cross-validation on 58 pairs gives coarse-grained, high-variance error estimates.
- BSIF performed close to chance alone (43.1% D-EER) despite contributing to the best fused score, which we attribute to the filter-bank substitution (Adaptation #1): fold-local ICA on a few dozen bona fide images is a much weaker texture prior than filters learned on 50,000 diverse natural-image patches.
- LBP and HOG carried almost all of the usable signal (~19% D-EER each); fusing all three still beat either stream alone, showing the ensemble/sum-fusion idea holds even when one stream is individually weak.
- Our attack set only contains $\alpha=0.5$ morphs from a single morphing pipeline (our own Part 1 Delaunay/warp code); a detector trained here is not shown to generalize to other morphing tools or α values, mirroring the generalization gap this literature repeatedly flags as an open problem.

7. Neural Network Fusion Layer

The paper combines the three descriptor-stream scores with a fixed, hand-picked sum rule. We additionally replace that fixed rule with the simplest possible neural network -- a single trainable neuron (logistic regression) -- that learns how much to weight each stream instead: $z = w_{\text{lbp}} \cdot \text{lbp} + w_{\text{hog}} \cdot \text{hog} + w_{\text{bsif}} \cdot \text{bsif} + b$, $p = \text{sigmoid}(z)$, trained by full-batch gradient descent on the binary cross-entropy loss. Its 3 inputs are the same z-scored, out-of-fold CRC scores from Section 5/6 -- already leakage-free -- so this experiment only needed one further pair-disjoint 80/20 train/test split (138 training images / 36 test images) to fit and evaluate the neuron itself.

src/mad_nn.py -- the fusion neuron:

```
class LogisticNeuron:
    def fit(self, X, y):
        # X = [lbp_score, hog_score, bsif_score] per image
        for _ in range(epochs):
            p = sigmoid(Xn @ self.weights + self.bias)
            loss = -mean(y*log(p) + (1-y)*log(1-p)) # binary cross-entropy
            self.weights -= lr * Xn.T @ (p - y) / n # gradient descent
            self.bias -= lr * mean(p - y)
```

Metric	Value
Training accuracy	88.41%
Test accuracy	88.89%
Final training cross-entropy loss	0.3996
Weight - LBP stream	3.617
Weight - HOG stream	4.730
Weight - BSIF stream	-0.478
Bias	-2.465

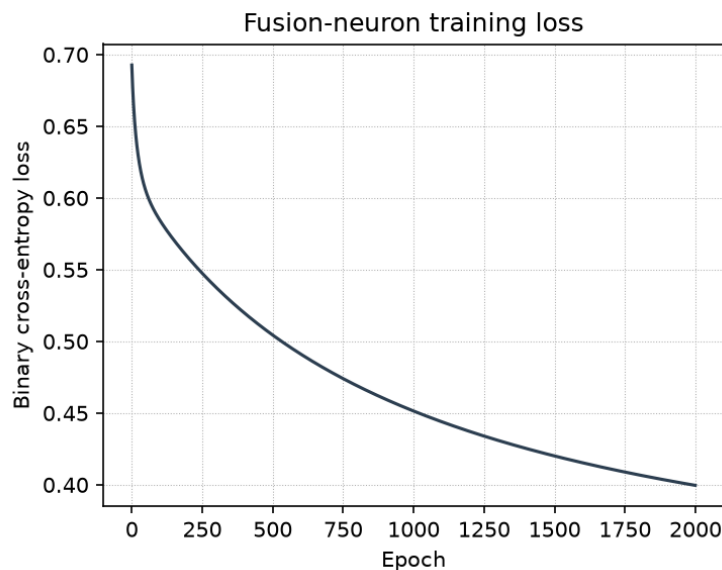


Figure 2: Binary cross-entropy training loss of the fusion neuron over 2000 gradient-descent epochs (learning rate 0.1).

The learned weights corroborate Section 6's finding purely from the training data, with no hand-tuning: LBP and HOG get large positive weights (the neuron relies on them heavily), while BSIF's weight is small and negative -- the network learned on its own that the BSIF stream carries little useful signal, consistent with its near-chance 43.1% standalone D-EER.

8. Conclusion

We reimplemented the ensemble-of-features S-MAD pipeline from Venkatesh et al. (FUSION 2020) end-to-end on our own bona fide/morph dataset from Part 1: color-space expansion, Laplacian-pyramid scale space, LBP/HOG/BSIF descriptors, per-stream P-CRC classification, and sum-rule fusion, evaluated with the ISO/IEC 30107-3 metrics the field standardizes on. The pipeline detects our own morphs meaningfully better than chance (D-EER 17.2%), with the expected accuracy gap to the original paper explained by dataset scale and our BSIF filter-bank substitution, both documented above. We further trained a small neural network (a single logistic-regression neuron) to learn the stream-fusion weights directly from data instead of a fixed sum rule, reaching 88.9% test accuracy and independently confirming, through its learned weights, which descriptor streams actually carry signal.